Formation and Evolution of Cataclysmic Variable Stars

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Abstract

This essay will cover the current understanding of formation and evolution of Cataclysmic Variables (CVs), by discussing the basics of binary stars, main sequence star and white dwarf formation, accretion, Roche lobe overflow, and mass transfer stability and angular momentum loss. CVs are important in the astrophysics field due to how they have helped the understanding of the aforementioned processes, like mass transfer, and others like accretion, as well as helping to develop theories behind the formation of phenomena like Type Ia Supernovae. This essay is aimed at an audience of 2nd year astrophysicists.

1 Introduction to Cataclysmic Variable Stars

A binary system contains two stars or objects that are gravitationally bound, and orbit a barycentric mass, with a large orbital period. In a binary star system, the more massive or larger of the two stars is known as the primary star, and the less massive or smaller of the two stars is the secondary star.

In terms of a CV, the system contains a white dwarf primary star, and a main sequence secondary star, which orbit in a short, close, period. The name cataclysmic variable, comes from the Greek *kataklysmos*, meaning flood or storm (Hack et al. 1993). The closeness of these systems causes mass to transfer, and then accretion of that mass from the main sequence star to the white dwarf, which can also cause x-ray emission and sudden bright flashes (novae).

2 Formation of Cataclysmic Variable Stars

For completeness here, the objects in the system will be explained, along with a basis on how they are formed.

2.1 Objects in the CV binary

2.1.1 Main Sequence Star

A main sequence star, is a star that has a mass of range 0.1 to 200 times that of our sun ($M_{\odot}$ = 1.989x10^{30} kg) (Tillman & Biggs 2022), and in its primary hydrogen burning stage of nuclear fusion. This stage is known as the main sequence of a stars lifetime. Below is a Hertzsprung-Russell diagram, which plots each stage of a stars lifetime using the luminosities and temperatures of the stars.

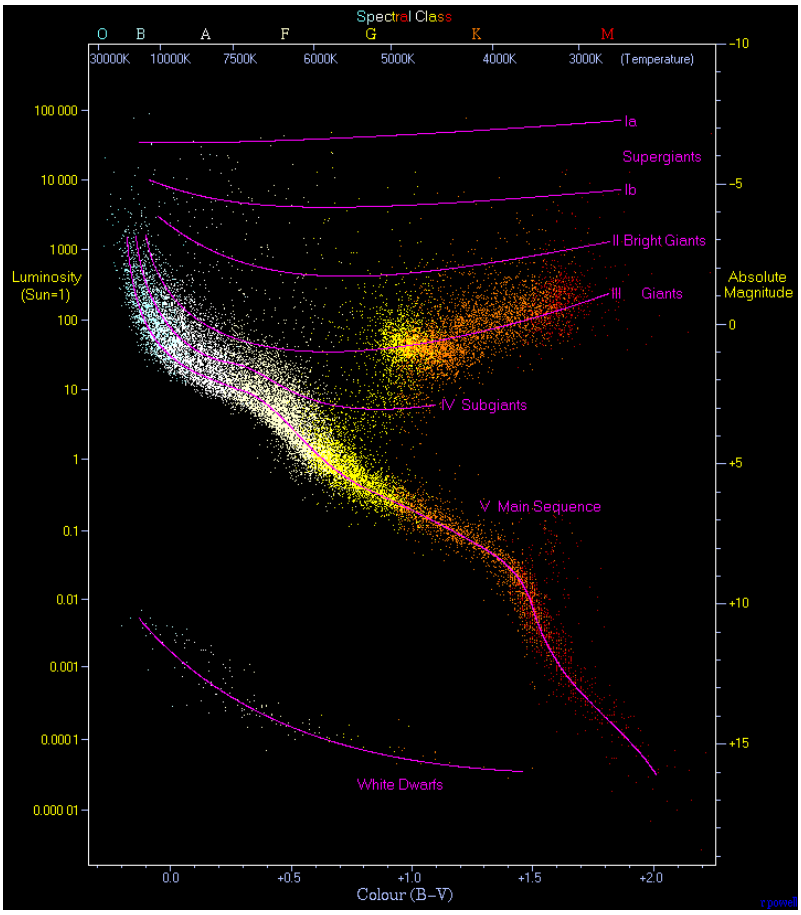

Figure 1: An example H-R diagram taken from Powell (2006), compiling 22000 stars from the Hipparcos catalogue.

These low mass stars are the main suppliers of mass during mass transfer, and

essentially drive the evolution of the CV system.

2.1.2 White dwarf Star

One of the possible endings of a star's life is a white dwarf (WD), which generally occurs in low mass stars between $0.07M_\odot$ and $8M_\odot$ (Fontaine et al. 2001), where the star does not have enough mass once it has evolved off the main sequence to fuse heavier elements than carbon in its core. This causes the stars outer shell to shed into a planetary nebulae, leaving behind a dense core - the white dwarf. The WD has a high surface gravity, which enables the star to undergo an efficient level of accretion.

2.2 Mass transfer process in a CV binary

2.2.1 Roche Lobe

Named after Édouard Roche, the Roche lobe, r_L, is the region around a star in a binary system, which limits how far any orbiting mass can be gravitationally bound to the respective star, which itself is limited by the Lagrangian Point, L_1. L_1 is bound to the stars gravitational potential, Φ, and gives a teardrop shape to the Roche lobe. A diagram of the Roche lobe and Lagrangian point of a main sequence star and white dwarf can be seen below.

The equation for calculating the size of a stars Roche lobe can be found below. Taken from Eggleton (1983)

$$r_L = \frac{0.49q^{2/3}}{0.6q^{2/3} + ln(1 + q^{1/3})}, 0 < q < \infty \tag{1}$$

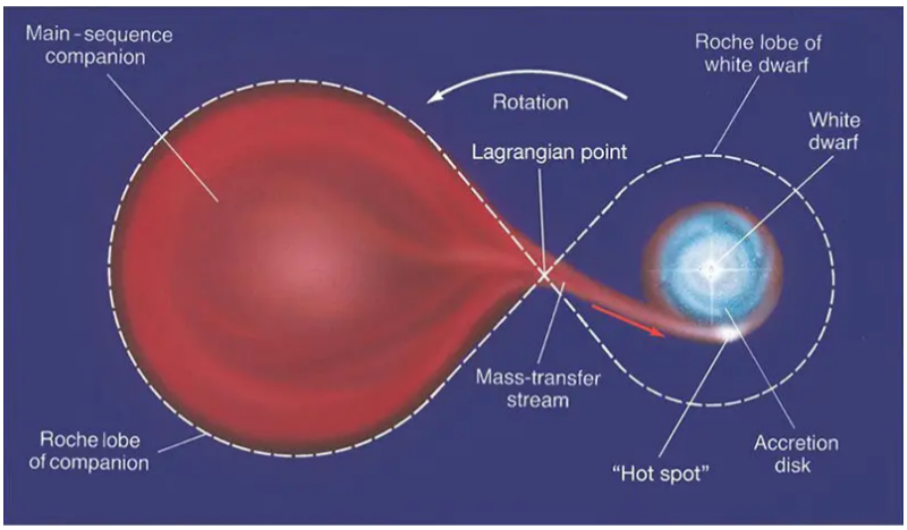

Figure 2: A diagram showing the Roche lobe and Lagrangian point on a CV system, taken from Ashworth (2019)

where q is the mass ratio of the two stars;

$$q = \frac{M_1}{M_2} \tag{2}$$

where M_1 is the mass of the primary star and M_2 is the mass of the secondary star.

Equation (1) shows that for a given mass ratio q, as it tends to an extreme value, the secondary main sequence star's Roche lobe shrinks, which creates unstable mass transfer.

Depending on if the Roche lobe of the stars have been filled, determines the type of CV they are; a detached CV: where both of the stars are within their Roche lobe, a semi-detached CV: where the secondary star has filled its Roche lobe, but the primary remains within its own, and a contact CV: where both stars have filled their Roche Lobe.

2.2.2 Mass Transfer

Mass transfer is a process which allows mass that is gravitationally bound to the secondary star, now known as a donor star (DS), fills the stars Roche lobe. When the mass passes L_1, it flows into and becomes gravitationally bound to the white dwarf star by accretion. This process is known as Roche Lobe Overflow.

The mass loss rate from the DS, taken from Warner (1995), can be seen below:

$$\dot{M} = Q\rho_{L_1}c_s \tag{3}$$

where Q is the effective cross-section of the stream of mass, ρ_{L_1} is the density at the Lagrangian point over Q, L_1, and c_s is the averaged isothermal sound velocity.

2.3 Accretion disk in a CV binary

When mass is transferred to the primary white dwarf star via Roche lobe overflow, the white dwarf undergoes accretion, the growth of mass of the white dwarf due to its own gravitational attraction. By accreting the mass from the DS, the white dwarf also takes in the angular momentum and energy the mass had when being transferred. This causes the mass to spiral in an inward motion while orbiting the gravitational field of the white dwarf, causing a flow of mass known as an Accretion Disk. The inward motion of material causes angular momentum to spread outwards, allowing accretion onto the WD. A flow diagram of the creation of an accretion disk can be seen in the figure below.

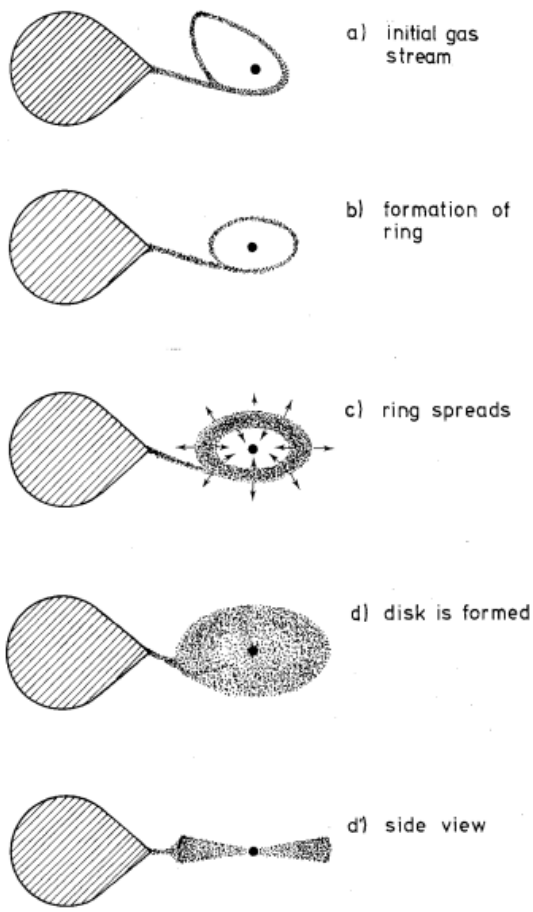

Figure 3: Example formation of an accretion disk in a CV system, taken from Verbunt (1982)

3 Evolution of Cataclysmic Variable Stars

3.1 Angular Momentum Loss

Angular Momentum Loss, AML, is thought to be the driving force behind mass transfer between binary systems, and in this case, CVs, also, AML also dictates how quickly CVs can evolve. This section will take a look at the types of AML known, and then discuss potential end of life events for a CV in terms of a Type Ia Supernovae. The two preimary types of AML mechanisms that will be discussed are Magnetic Braking, and Gravitational Wave Radiation.

3.1.1 Magnetic Braking

The main driving mechanism, and the reason for the highest amount of AML in a CV, is Magnetic Braking (MB). MB is the dominant form of AML for CVs with orbital periods, P_{orb}, above 3 hours. MB occurs when a DS's stellar wind gets blown towards its Alfvén radius due to its magnetic field from rotation and internal convection (Barraza-Jorquera et al. 2025). The stellar wind gets attached to the magnetic field lines of the DS. Since the DS is rotating, the magnetic field also rotates, causing the stuck wind to rotate in turn. This complete rotational motion creates an arm, as the motion causes gas to spread further into space, and rotate at a speed equal to the DSs rotational speed. When the stellar wind becomes too powerful for the DSs magnetic field to hold, it breaks free from the arm and spreads past the Alfvẽn radius, which removes angular momentum from the DS, and the CV system as a whole. Because the DS is tidally locked to the WD, the rotational period (P_{rot} is equal to the P_{orb}. This braking motion because of the AML, brakes the entire orbit. The AML then makes the two stars spiral towards each other, which in turn, increases the rate of mass transfer (Verbunt & Zwaan 1981). MB does stop occuring if the P_{orb} is less than 3 hours (Knigge et al. 2011).

3.1.2 Gravitational Wave Radiation

Compared to MB, Gravitational Wave Radiation (GWR), is a constantly occuring form of AML. However, compared to MB, GWR is smaller rate of AML and tends to be dominated by MB in P_{orb} over 3 hours. But as mentioned previously, MB stops at P_{orb} equal to 3 hours, allowing for GWR to dominate in periods less than that, generally in P_{orb} less than 2 hours (Knigge et al. 2011). The gap between 2-3 hours, known as the Orbital Period Gap, will be discussed later. Since the DS

and WD stars have a gravitational field, they constantly emit gravitational waves outwards from the core. These waves are able to carry angular momentum as the stars orbit, losing momentum from the system. Current research believes that the GWR emitted from CV systems (and WD stars in particular) could be useful target candidates for LISA (see Belloni & Schreiber (2023) and Amaro-Seoane et al. (2023) for a more in depth discussion).

3.1.3 The Orbital Period Gap

As mentioned previously, the gap between the end of MB at around 3 hour P_{orb}, and the 2 hour P_{orb} end of GWR, is known as the Orbital Period Gap (Schreiber et al. 2024, Shao & Li 2012). This gap is a supposed dead zone for CVs, due to the disruption of AML and mass transfer. MB effectively shuts off, as noted by Knigge et al. (2011), towards the 3 hour P_{orb} point, at which the DS becomes convective and the high rate of mass transfer stops. The DS falls out of thermal equilibrium and appears to be larger in size than its mass should allow it to be - should it be keeping the same density it was. After MB stops, DS contracts, losing contact with its Roche lobe. When the Roche lobe catches up to the size of the radius of the DS at around 2 hour P_{orb}, mass transfer resumes, with the AML being pushed by GWR instead of MB.

3.2 Evolution to a Type Ia Supernova

3.2.1 Classical Novae

A nova, is an outburst of sudden bright light. The common types of novae are; Classical, Recurring and Dwarf, all of which are known to originate from CV binaries. Classical novae are formed when mostly hydrogen gas is accreted by the

WD during mass transfer, gathers around the surface of the WD before forming into an accretion disk. The hydrogen mass is then heated by the internal temperature of the WD's surface and reaches a critical temperature, T_{crit}, which leads to the ignition of thermal runaway fusion (See Cowan et al. (1982) for an in detail explanation of this effect). A sudden increase in energy causes an expulsion of the surrounding atmosphere, which creates the novae outbursts, in this case, Classical novae. WD stars can have multiple classical novae occur over time, when more hydrogen gets accreted on the surface of the WD from the DS. This repetition of classical novae is now classified as Recurring novae.

3.2.2 Type Ia Supernovae

In this subsection, an outline of the current understanding and interpretation of the formation mechanism of a Type Ia Supernova (SN Ia) will be given, as there is no definitive and accepted factual mechanism.

When a WD accretes matter from the DS, it has a slight chance to pass the Chandrasekhar limit ($M_{chan} = 1.44M_{\odot}$, Chandrasekhar (1931)), meaning that the electron degeneracy pressure overtakes the gravitational potential of the WD, and it collapses to form a neutron star. However, in terms of a SN Ia, the current theory is that the WD cannot pass the Chandrasekhar limit, and the collapse into a neutron star fails. An increase in pressure and density raises the temperature of the WD, as it reaches around 99% of the Chandrasekhar limit, putting the star into a convection period which lasts around 1000 years. This period is known as the simmering phase (Ruiter & Seitenzahl 2024). After the simmering phase, a deflagration flame front is created, which is powered and formed by the fusion of carbon in the core of the WD from thermonuclear runaway (Hoyle & Fowler 1960). This fusion causes the WD temperature to increase, forming heavier elements like

Iron, and the energy release from this fusion process overcomes the WD gravitational energy. This energy release manifests itself as a SN Ia, and the WD becomes unbound, and leaves no remnant - like a neutron star or black hole. This underlying theory for the formation of a Type Ia supernova from a single degenerative WD binary progenitor can be seen in Whelan & Iben (1973). In the current field of CVs and WD binary systems, the formation of SN Ia's is debated, with some researchers and academics arguing that SN Ia's are formed via a single-degenerate progenitor model like the aforementioned Whelan & Iben (1973), whilst others like Webbink (1984) produced theories of a double-degenerate merger model which shows a model of a double WD star binary merging to form a SN Ia.

4 Recent CV research and findings

In an attempt to further study the Orbital Period Gap, Larsen & MacDonald (2025) modelled the Convective Kissing Instability (CKI) for CV system stellar evolution. CKI (see van Saders & Pinsonneault (2012) for an in depth overview) occurs in stellar evolution models with a mass around 1.35 $M_{\odot}$, that is a fully convective transition mass. The CKI is driven by non-equilibrium helium burning, that causes a merger of convective zones. This merger causes changes in the radii and luminosity of the system - referred to by both Larsen & MacDonald (2025) and van Saders & Pinsonneault (2012) as "glitches". In the original paper, van Saders & Pinsonneault (2012) suggested that these glitches could lead to the detachment of a CV, and an enhanced mass loss rate. Larsen & MacDonald (2025) found that in CVs that evolve via MB and GWR above the Orbital Period Gap, CKI has no effect, and that CKI does not occur and/or is halted when mass transfer starts in the system. Continuing, Larsen & MacDonald (2025) stated if AML is driven by GWR only, CKI occurs, and causes a small detachment phase which produces a small period gap of a few minutes, where Larsen & MacDonald (2025) suggested that this could be more evidence for polars (see Warner (1995) for an overview). Finally, the results from Larsen & MacDonald (2025) indicate that CKI can cause a small detachment phase in around three hour orbital period systems, however, this gap caused by CKI, is not sufficient enough to explain the standard 2-3 hour orbital period gap.

Joshi et al. (2025) looked at three CVs; CRTS J110014.7+131552 (J1100), SDSS J093537.46+161950.8 (J0935) and [PK2008] HalphaJ130559 (J130559), to attempt to detect and analyse superhumps (see Vogt (1974) for the first detection of superhumps/superoutbursts, and Montgomery (2012) for an overview on disk preces-

sion in CVs).

4.1 J1100

Joshi et al. (2025) confirmed J1100 as a SU UMa-type dwarf nova, by identifying a 14.5 day prolonged outburst and a decreasing superhump amplitude as the outburst fades, from the TESS lightcurve (see Ricker et al. (2014) for an overview on the TESS mission before launch) which is a characteristic of a superoutburst of a SU UMa-type star. Variations identified in the superhump period suggested to Joshi et al. (2025) that there are changes in the accretion disk radius throughout the superburst, which supports the thermal tidal instability model (Osaki 1989). The TESS lightcurve for J1100 can be seen below, in the top panel.

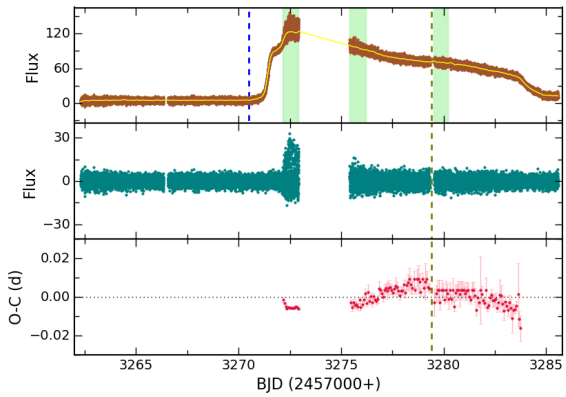

Figure 4: J1100's lightcurve from TESS data, which displays a superoutburst. Figure taken from Joshi et al. (2025).

4.2 J0935

J0935 was identified as a high mass transfer CV below the Orbital Period Gap. TESS observations found evidence of a positive superhump of 0.06584 days, and a corresponding disc-precession period of 2.36 days, which was not identified in previous works like Southworth, J. et al. (2015). J0935 was found to have a short orbital period and a positive superhump, but lacks the outburst activity

during long term observations. This evidence suggested to Joshi et al. (2025) that the system is not an SU UMa-type dwarf nova, but could be a high mass transfer CV under the 2-3 hour period gap, which led to the idea of J0935 being an intermediate polar.

4.3 J130559

Lastly, J130559 was identified as a candidate for a tilted disc. The TESS data for this system gave a new averaged periodicity of 3.83 days as well as displaying periodicities of 0.15092 and 0.14517 days (solving what was previously an ambiguity in earlier period determinations). Importantly, the 0.159092 day period signal confirmed the systems orbital period. The 0.14517 day and 3.83 day period signals were identified as the negative superhump and disc-precession periods respectively, which if confirmed to relate to the corresponding parameter in future research, Joshi et al. (2025) claims these values indicate retrograde precession of a tilted accretion disk, and an interaction between the tilted disk and the secondary stream. This interaction would be a new example for tilted disk precession studies in CVs.

5 Conclusion

In conclusion, Cataclysmic Variable systems contain a white dwarf primary star, and a main sequence secondary star, which orbit in a short, close, period - and are ideal systems to act as laboratories for in depth study of mass transfer processes and respective angular momentum loss. The mass transfer process is initiated by Roche lobe overflow, and during this process, angular momentum is lost via magnetic braking and gravitational wave radiation. These mechanisms create the 2-3 hour orbital period gap. While the formation of a Type Ia supernova has a general accepted mechanism, the progression of a CV system into a SN Ia has two main progenitor theories; a single white dwarf progenitor binary, and a double white dwarf merger binary. Understanding and researching the evolution of CV systems is beneficial, as it aides the understanding of accretion physics, processes like mass transfer and novae outbursts, as well as possible reasons for the orbital period gap.

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